

04_Final_Analysis

Contents

Introduction	1
Table 1	2
Sequential Models	2
Model 1: Crude Model	2
Interpretation	4
Model 2: Demographics	4
Interpretation	4
Model 3: Risk Factors	5
Interpretation	6
Model 4: Parental Monitoring Knowledge	6
Interpretation	7
Sensitivity Analysis	7
Table 1	7
Mod 1	8
Mod 2	9
Mod 3	9

Introduction

This script contains the final output for the final analyses conducted for aim 1, which were previously generated in the file `Scripts/03_Analysis.Rmd`. The output in this document will cover:

1. Descriptive Statistics of the sample stratified by age group (**Table 1**)
2. **Sequential Models** used to assess the association between age group and reported driving after drinking.

Table 1

We will compare our two age groups among our nationally representative sample (weighting applied) across variables of interest, including demographics, risk factors, and parental monitoring.

Note In analyses from the 03_Analysis.rmd file we also compared other variables, such as parenting styles and risk categories for physical environment and family/home risk, but these are excluded from this output due to them ultimately not being informative for our models and/or having too much missing data.

```
design_table_1 <- svydesign(  
  ids = ~id,  
  weights = ~weight,  
  data = study_dat  
)  
  
num_vars_v2 <- c(  
  "num_days_drink_02"  
)  
  
fac_vars_cat <- c(  
  "drive_after_drink", "sex",  
  "ride_drink_2002", "binge", "marijuana",  
  "license", "smoker", "race_eth", "education", "income",  
  "avg_pmk_cat"  
)  
  
tbl1_cat <- tbl_svysummary(  
  design_table_1,  
  by = age_group,  
  include = c(all_of(num_vars_v2), all_of(fac_vars_cat)),  
  type = list(  
    binge ~ "categorical",  
    marijuana ~ "categorical",  
    ride_drink_2002 ~ "categorical"  
  ),  
  statistic = list(  
    all_continuous() ~ "{mean} ({sd})",  
    all_categorical() ~ "{n} ({p}%)"  
  )  
) %>%  
  add_p()  
  
tbl1_cat
```

Sequential Models

Model 1: Crude Model

This model evaluates the **crude association** between age group and driving after drinking, applying the weighting and survey design variables as needed to make our sample nationally representative.

We also assess the model for dispersion to make sure there is no overdispersion, warranting the use of a quasi-binomial specification; no overdispersion is indicated, so we proceed with using a binomial family.

```
mod1 <- svyglm(
  drive_after_drink ~ age_group,
  design = design,
  family = binomial()
)
```

```
summary(mod1)
```

```
##
## Call:
## svyglm(formula = drive_after_drink ~ age_group, design = design,
##       family = binomial())
##
## Survey design:
## svydesign(ids = ~id, weights = ~weight, data = study_dat)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -2.69857    0.06637  -40.66  <2e-16 ***
## age_group21-24  0.93103    0.08907   10.45  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1.000132)
##
## Number of Fisher Scoring iterations: 5
```

```
summary(mod1)$dispersion
```

```
##      variance      SE
## [1,]  1.0001 0.0388
```

```
tidy(mod1, conf.int = TRUE) %>%
  mutate(
    OR = exp(estimate),
    CI_low = exp(conf.low),
    CI_high = exp(conf.high),
    p_value = case_when(
      p.value < 0.001 ~ "< .001",
      TRUE ~ sprintf("%.2f", p.value)
    )
  ) %>%
  select(term, OR, CI_low, CI_high, p_value)
```

```
## # A tibble: 2 x 5
##   term          OR CI_low CI_high p_value
##   <chr>      <dbl> <dbl>  <dbl> <chr>
## 1 (Intercept)  0.0673 0.0591  0.0767 < .001
## 2 age_group21-24 2.54   2.13   3.02   < .001
```

Interpretation

In the survey-weighted unadjusted logistic regression model, individuals aged 21-24 years old had significantly higher odds (OR = 2.5, 95% CI: 2.13 to 3.02, $p < .001$) of reporting driving after drinking compared with individuals >21 years.

Model 2: Demographics

Next, we are going to add some demographic characteristics. We are going to start with sex, race/ethnicity, and income.

```
mod2 <- svyglm(  
  drive_after_drink ~ age_group + sex + race_eth + income,  
  design = design,  
  family = binomial()  
)  
  
tidy(mod2, conf.int = TRUE) %>%  
  mutate(  
    OR = round(exp(estimate),2),  
    CI_low = round(exp(conf.low),2),  
    CI_high = round(exp(conf.high),2),  
    p_value = case_when(  
      p.value < 0.001 ~ "< .001",  
      TRUE ~ sprintf("%.2f", p.value)  
    )  
  ) %>%  
  select(term, OR, CI_low, CI_high, p_value)
```

```
## # A tibble: 10 x 5  
##   term                                OR CI_low CI_high p_value  
##   <chr>                            <dbl> <dbl> <dbl> <chr>  
## 1 (Intercept)                     0.04  0.03  0.06 < .001  
## 2 age_group21-24                   2.31  1.9   2.82 < .001  
## 3 sexMale                           1.85  1.51  2.27 < .001  
## 4 race_ethBlack                     0.47  0.35  0.63 < .001  
## 5 race_ethHispanic                  0.76  0.58  0.98 0.04  
## 6 race_ethMixed Race (Non-Hispanic) 1.16  0.51  2.66 0.72  
## 7 income$10,001 - $30,000           1.17  0.82  1.67 0.39  
## 8 income$30,001 - $50,000           1.3   0.9   1.87 0.16  
## 9 income$50,001 - $80,000           1.24  0.86  1.79 0.24  
## 10 income>$80,000                   1.52  1.08  2.15 0.02
```

Interpretation

After adjusting for sex, race/ethnicity, and income, individuals aged 21-24 had 2.3x the odds of reporting driving after drinking (95% CI: 1.89 to 2.82, $p < .001$).

Independent effects

Females had significantly reduced odds of driving after drinking compared to males (OR = 0.54, $p < .001$), holding age group, race/ethnicity, and income constant.

Compared to Non-Hispanic Non-Blacks (aka White), Blacks had significantly reduced odds (OR = 0.47, $p < .001$), Hispanics had significantly reduced odds (OR = 0.76, $p = 0.04$) of driving after drinking. Mixed Race (Non-Hispanic) individuals had no significant odds (OR = 1.24, $p = 0.72$).

For income, there was no significant odds ratios for the income groups compared to the lowest income group (<\$10k) except for the largest income group (>\$80k). We also see that the OR for each income group increases as income goes up.

Model 3: Risk Factors

Next, we are going to add some risk/protective factors around driving and drinking, paying special attention to the variables that popped as having a statistically significant association with our exposure. While education pops up, we will not be including it due to its colinearity with income. The only variables that were statistically significant in our table 1 for having different distributions among our groups were the number of days drinking and riding with a drinking driver, so we will include those two and see what we get.

```
mod_3 <- svyglm(
  drive_after_drink ~ age_group + sex + race_eth + income +
    num_days_drink_02 + ride_drink_2002 ,
  design = design,
  family = binomial()
)

tidy(mod_3, conf.int = TRUE) %>%
  mutate(
    OR = round(exp(estimate),2),
    CI_low = round(exp(conf.low),2),
    CI_high = round(exp(conf.high),2),
    p_value = case_when(
      p.value < 0.001 ~ "< .001",
      TRUE ~ sprintf("%.2f", p.value)
    )
  ) %>%
  select(term, OR, CI_low, CI_high, p_value)
```

```
## # A tibble: 12 x 5
##   term                                OR CI_low CI_high p_value
##   <chr>                             <dbl> <dbl> <dbl> <chr>
## 1 (Intercept)                       0.02  0.01  0.03 < .001
## 2 age_group21-24                     1.78  1.42  2.23 < .001
## 3 sexMale                           1.55  1.23  1.96 < .001
## 4 race_ethBlack                      0.59  0.42  0.84 0.00
## 5 race_ethHispanic                   0.9   0.67  1.21 0.47
## 6 race_ethMixed Race (Non-Hispanic) 1.14  0.35  3.7  0.82
## 7 income$10,001 - $30,000            1.27  0.85  1.9  0.24
## 8 income$30,001 - $50,000            1.43  0.95  2.15 0.09
## 9 income$50,001 - $80,000            1.31  0.88  1.96 0.18
## 10 income>$80,000                    1.36  0.93  1.99 0.12
## 11 num_days_drink_02                  1.08  1.07  1.1  < .001
## 12 ride_drink_2002Yes                 5.65  4.49  7.1  < .001
```

Interpretation

After adjusting for sex, race/ethnicity, income, number of days drinking, whether they rode with a drinking driver, whether they have a driver's license, the number of days they drank, whether they reported binge drinking over the past 30 days, and whether they used marijuana in the past 30 days, **individuals aged 21-24 had 1.8x the odds of reporting driving after drinking (95% CI: 1.42 to 2.23, $p < .001$).**

Independent effects

Controlling for all other variables, each one unit increase in the number of days drinking was associated with a 1.08x increase in the odds of driving after drinking ($p = .01$).

Riding with a drinking driver greatly increased the odds of driving after drinking, controlling for all other variables (OR = 5.65, $p < .001$).

Model 4: Parental Monitoring Knowledge

Lastly, we are going to add our parental monitoring knowledge variable. This is categorized the same way as it is in the NLSY97, with 4 levels of parental monitoring (0 to 4, 5 to 8, 9 to 12, and 13 to 16). The lowest level of parental monitoring is the reference level, so all other groups are comparing to individuals whose parents monitor them/have knowledge of their whereabouts and actions the least.

Note that this variable is an average of both residential mothers and fathers from the first 4 years of the survey collection.

```
mod_4 <- svyglm(
  drive_after_drink ~ age_group + sex + race_eth + income +
    num_days_drink_02 + ride_drink_2002 +
    avg_pmk_cat,
  design = design,
  family = binomial()
)

tidy(mod_4, conf.int = TRUE) %>%
  mutate(
    OR = round(exp(estimate),2),
    CI_low = round(exp(conf.low),2),
    CI_high = round(exp(conf.high),2),
    p_value = case_when(
      p.value < 0.001 ~ "< .001",
      TRUE ~ sprintf("%.2f", p.value)
    )
  ) %>%
  select(term, OR, CI_low, CI_high, p_value)
```

```
## # A tibble: 15 x 5
##   term                                OR CI_low CI_high p_value
##   <chr>                             <dbl> <dbl> <dbl> <chr>
## 1 (Intercept)                       0.02  0.01  0.05 < .001
## 2 age_group21-24                     3.08  1.12  8.45 0.03
## 3 sexMale                           1.39  0.94  2.07 0.10
## 4 race_ethBlack                     0.94  0.57  1.55 0.81
## 5 race_ethHispanic                   0.81  0.47  1.38 0.44
## 6 race_ethMixed Race (Non-Hispanic) 1.24  0.29  5.26 0.77
## 7 income$10,001 - $30,000           0.85  0.41  1.75 0.65
```

## 8	income\$30,001 - \$50,000	1.62	0.81	3.24	0.17
## 9	income\$50,001 - \$80,000	1.67	0.89	3.13	0.11
## 10	income>\$80,000	1.34	0.74	2.45	0.34
## 11	num_days_drink_02	1.09	1.06	1.11	< .001
## 12	ride_drink_2002Yes	6.43	4.37	9.47	< .001
## 13	avg_pmk_cat13 T0 16	1.2	0.33	4.33	0.78
## 14	avg_pmk_cat5 T0 8	1.28	0.51	3.25	0.60
## 15	avg_pmk_cat9 T0 12	0.85	0.33	2.18	0.73

Interpretation

Holding all other covariates constant, **individuals aged 21-24 had 3.08x the odds of reporting driving after drinking (95% CI: 1.12 to 8.45, $p = .03$).**

Note that the parental monitoring knowledge variable contains a lot of missing data, and may be hard to use for the purpose of this study, as parental monitoring knowledge is likely to be very dependent on age and not necessarily impact driving while drinking outcomes in a cross-sectional manner (it may serve better in a longitudinal study to see how parental monitoring affects drinking and driving trajectories).

Sensitivity Analysis

Licensure popped as a variable that was significantly associated with age group and our outcome in early data explorations. We are going to conduct a brief sensitivity analysis to see how our effect estimates may or may not change after excluding individuals who did not hold a drivers license from the study.

First, let's check and see what our sample size will become after excluding them and subset the data:

```
study_dat %>%
  group_by(license) %>%
  summarize(n())
```

```
## # A tibble: 3 x 2
##   license   'n()'
##   <fct>     <int>
## 1 Unlicensed 1788
## 2 Licensed   5783
## 3 <NA>       27
```

Once we remove the 1,788 unlicensed individuals and 27 NA's, we are left with 5,783 study participants. Let's apply our weighting and construct a table 1.

Table 1

```
design_table_1_sensitivity <- svydesign(
  ids = ~id,
  weights = ~weight,
  data = license_dat
)

num_vars_v2 <- c(
```

```

    "num_days_drink_02"
  )

  fac_vars_cat <- c(
    "drive_after_drink", "sex",
    "ride_drink_2002", "binge", "marijuana",
    "license", "smoker", "race_eth", "education", "income",
    "avg_pmk_cat"
  )

  tbl1_sens <- tbl_svysummary(
    design_table_1_sensitivity,
    by = age_group,
    include = c(all_of(num_vars_v2), all_of(fac_vars_cat)),
    type = list(
      binge ~ "categorical",
      marijuana ~ "categorical",
      ride_drink_2002 ~ "categorical"
    ),
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} ({p}%)"
    )
  ) %>%
  add_p()

tbl1_sens

```

Looks like we now have 15,353,418 participants after applying weights.

We will proceed with the sequential models, and include their output below, with comments on whether anything substantially changed between the models.

Mod 1

```

mod1_sens <- svyglm(
  drive_after_drink ~ age_group,
  design = design,
  family = binomial()
)

tidy(mod1_sens, conf.int = TRUE) %>%
  mutate(
    OR = exp(estimate),
    CI_low = exp(conf.low),
    CI_high = exp(conf.high),
    p_value = case_when(
      p.value < 0.001 ~ "< .001",
      TRUE ~ sprintf("%.2f", p.value)
    )
  )

```

```
) %>%
select(term, OR, CI_low, CI_high, p_value)
```

```
## # A tibble: 2 x 5
##   term          OR CI_low CI_high p_value
##   <chr>      <dbl> <dbl>  <dbl> <chr>
## 1 (Intercept) 0.0775 0.0675 0.0890 < .001
## 2 age_group21-24 2.48 2.06 2.98 < .001
```

The odds ratio attenuated very slightly from 2.54 to 2.48; no other change.

Mod 2

```
mod2_sens <- svyglm(
  drive_after_drink ~ age_group + sex + race_eth + income,
  design = design,
  family = binomial()
)

tidy(mod2_sens, conf.int = TRUE) %>%
  mutate(
    OR = round(exp(estimate),2),
    CI_low = round(exp(conf.low),2),
    CI_high = round(exp(conf.high),2),
    p_value = case_when(
      p.value < 0.001 ~ "< .001",
      TRUE ~ sprintf("%.2f", p.value)
    )
  ) %>%
  select(term, OR, CI_low, CI_high, p_value)
```

```
## # A tibble: 10 x 5
##   term          OR CI_low CI_high p_value
##   <chr>      <dbl> <dbl>  <dbl> <chr>
## 1 (Intercept) 0.06 0.04 0.08 < .001
## 2 age_group21-24 2.3 1.87 2.84 < .001
## 3 sexMale      1.84 1.49 2.27 < .001
## 4 race_ethBlack 0.51 0.37 0.7 < .001
## 5 race_ethHispanic 0.78 0.59 1.03 0.08
## 6 race_ethMixed Race (Non-Hispanic) 1.41 0.59 3.36 0.44
## 7 income$10,001 - $30,000 1 0.69 1.46 0.99
## 8 income$30,001 - $50,000 1.12 0.77 1.64 0.55
## 9 income$50,001 - $80,000 0.97 0.66 1.42 0.87
## 10 income>$80,000 1.14 0.8 1.63 0.45
```

No major changes.

Mod 3

```

mod_3_sens <- svyglm(
  drive_after_drink ~ age_group + sex + race_eth + income +
    num_days_drink_02 + ride_drink_2002,
  design = design,
  family = binomial()
)

tidy(mod_3_sens, conf.int = TRUE) %>%
  mutate(
    OR = round(exp(estimate),2),
    CI_low = round(exp(conf.low),2),
    CI_high = round(exp(conf.high),2),
    p_value = case_when(
      p.value < 0.001 ~ "< .001",
      TRUE ~ sprintf("%.2f", p.value)
    )
  ) %>%
  select(term, OR, CI_low, CI_high, p_value)

```

```

## # A tibble: 12 x 5
##   term                                OR CI_low CI_high p_value
##   <chr>                            <dbl> <dbl> <dbl> <chr>
## 1 (Intercept)                      0.02  0.02  0.04 < .001
## 2 age_group21-24                    1.69  1.33  2.14 < .001
## 3 sexMale                          1.62  1.27  2.07 < .001
## 4 race_ethBlack                     0.7   0.47  1.03 0.07
## 5 race_ethHispanic                  0.88  0.63  1.21 0.43
## 6 race_ethMixed Race (Non-Hispanic) 1.45  0.38  5.51 0.59
## 7 income$10,001 - $30,000           1.26  0.83  1.92 0.28
## 8 income$30,001 - $50,000           1.34  0.87  2.05 0.19
## 9 income$50,001 - $80,000           1.06  0.69  1.62 0.79
## 10 income>$80,000                   1.07  0.72  1.61 0.73
## 11 num_days_drink_02                 1.08  1.07  1.1  < .001
## 12 ride_drink_2002Yes                6.13  4.82  7.8  < .001

```

The odds ratio attenuated to 1.69, remains significant at $\sim .01$. Ride while drunk increased slightly to 4.55. Marijuana became even less significant.

```

mod_4_sens <- svyglm(
  drive_after_drink ~ age_group + sex + race_eth + income +
    num_days_drink_02 + ride_drink_2002 + avg_pmk_cat,
  design = design,
  family = binomial()
)

tidy(mod_4_sens, conf.int = TRUE) %>%
  mutate(
    OR = round(exp(estimate),2),
    CI_low = round(exp(conf.low),2),
    CI_high = round(exp(conf.high),2),
    p_value = case_when(
      p.value < 0.001 ~ "< .001",

```

```

    TRUE ~ sprintf("%.2f", p.value)
  )
) %>%
select(term, OR, CI_low, CI_high, p_value)

```

```

## # A tibble: 15 x 5
##   term                                OR CI_low CI_high p_value
##   <chr>                        <dbl> <dbl> <dbl> <chr>
## 1 (Intercept)                0.03  0.01  0.1 < .001
## 2 age_group21-24             2.82  0.98  8.1  0.05
## 3 sexMale                    1.52  1      2.32 0.05
## 4 race_ethBlack              1.14  0.65  2     0.65
## 5 race_ethHispanic           0.7   0.38  1.28 0.24
## 6 race_ethMixed Race (Non-Hispanic) 1.66  0.31  8.97 0.56
## 7 income$10,001 - $30,000    0.97  0.45  2.1  0.94
## 8 income$30,001 - $50,000    1.6   0.77  3.35 0.21
## 9 income$50,001 - $80,000    1.54  0.77  3.06 0.22
## 10 income>$80,000           1.06  0.55  2.05 0.86
## 11 num_days_drink_02         1.09  1.06  1.11 < .001
## 12 ride_drink_2002Yes       7.37  4.89 11.1 < .001
## 13 avg_pmk_cat13 T0 16      0.87  0.23  3.28 0.84
## 14 avg_pmk_cat5 T0 8        0.8   0.3   2.09 0.64
## 15 avg_pmk_cat9 T0 12       0.53  0.2   1.41 0.20

```

Overall, it does not appear as though excluding licensed drivers had a meaningful difference on our observed results.

Characteristic	Pre-21 N = 11,399,457 ¹	21-24 N = 7,467,581 ¹
num_days_drink_02	5 (6)	7 (7)
Unknown	3,693,689	1,466,108
drive_after_drink	718,825 (6.3%)	1,089,132 (15%)
sex		
Female	5,595,639 (49%)	3,640,357 (49%)
Male	5,803,818 (51%)	3,827,223 (51%)
ride_drink_2002		
No	9,629,601 (85%)	5,777,573 (77%)
Yes	1,756,177 (15%)	1,686,683 (23%)
Unknown	13,679	3,325
binge		
No	2,536,049 (39%)	2,112,386 (40%)
Yes	3,887,927 (61%)	3,183,135 (60%)
Unknown	4,975,481	2,172,059
marijuana		
No	738,167 (24%)	437,638 (24%)
Yes	2,357,626 (76%)	1,420,472 (76%)
Unknown	8,303,665	5,609,470
license		
Unlicensed	2,330,954 (20%)	1,116,074 (15%)
Licensed	9,041,840 (80%)	6,311,578 (85%)
Unknown	26,663	39,929
smoker		
No	145,106 (3.3%)	88,940 (2.8%)
≥ 1 Cig in Last 30 Days	4,232,734 (97%)	3,122,433 (97%)
Unknown	7,021,617	4,256,208
race_eth		
Non-Black / Non-Hispanic	8,141,262 (71%)	5,335,724 (71%)
Black	1,711,183 (15%)	1,076,810 (14%)
Hispanic	1,405,991 (12%)	971,180 (13%)
Mixed Race (Non-Hispanic)	141,020 (1.2%)	83,867 (1.1%)
Highest educational attainment		
Associate/Junior college (AA)	74,441 (0.9%)	206,152 (3.1%)
Bachelor's degree (BA, BS)	0 (0%)	330,188 (5.0%)
GED	511,966 (6.1%)	453,300 (6.8%)
High school diploma (Regular 12 year program)	7,817,675 (93%)	5,626,418 (85%)
Master's degree (MA, MS)	0 (0%)	964 (<0.1%)
PhD	0 (0%)	3,801 (<0.1%)
Unknown	2,995,375	846,757
income		
≤\$10,000	1,365,275 (17%)	981,430 (15%)
\$10,001 - \$30,000	1,514,446 (19%)	1,517,674 (24%)
\$30,001 - \$50,000	1,352,362 (17%)	1,216,585 (19%)
\$50,001 - \$80,000	1,621,927 (20%)	1,104,606 (17%)
>\$80,000	2,301,264 (28%)	1,516,127 (24%)
Unknown	3,244,184	1,131,159
avg_pmk_cat		
0 TO 4	472,601 (5.3%)	23,250 (19%)
13 TO 16	658,742 (7.5%)	6,730 (5.6%)
5 TO 8	3,353,309 (38%)	51,824 (43%)

Characteristic	Pre-21 N = 9,041,840 ¹	21-24 N = 6,311,578 ¹	p
num_days_drink_02	5 (6)	7 (7)	
Unknown	2,643,338	1,077,715	
drive_after_drink	650,213 (7.2%)	1,017,808 (16%)	
sex			
Female	4,414,585 (49%)	3,104,871 (49%)	
Male	4,627,255 (51%)	3,206,707 (51%)	
ride_drink_2002			
No	7,701,684 (85%)	4,857,090 (77%)	
Yes	1,330,840 (15%)	1,451,162 (23%)	
Unknown	9,317	3,325	
binge			
No	2,111,714 (39%)	1,832,601 (40%)	
Yes	3,247,350 (61%)	2,796,436 (60%)	
Unknown	3,682,776	1,682,541	
marijuana			
No	633,637 (26%)	374,386 (24%)	
Yes	1,840,633 (74%)	1,184,935 (76%)	
Unknown	6,567,570	4,752,256	
license			
Unlicensed	0 (0%)	0 (0%)	
Licensed	9,041,840 (100%)	6,311,578 (100%)	
smoker			
No	128,113 (3.8%)	79,820 (3.0%)	
≥ 1 Cig in Last 30 Days	3,220,230 (96%)	2,547,560 (97%)	
Unknown	5,693,497	3,684,197	
race_eth			
Non-Black / Non-Hispanic	7,050,802 (78%)	4,778,305 (76%)	
Black	978,313 (11%)	712,660 (11%)	
Hispanic	921,799 (10%)	753,257 (12%)	
Mixed Race (Non-Hispanic)	90,927 (1.0%)	67,355 (1.1%)	
Highest educational attainment			
Associate/Junior college (AA)	69,576 (1.0%)	195,233 (3.3%)	
Bachelor's degree (BA, BS)	0 (0%)	301,876 (5.2%)	
GED	370,131 (5.1%)	315,045 (5.4%)	
High school diploma (Regular 12 year program)	6,826,719 (94%)	5,027,144 (86%)	
Master's degree (MA, MS)	0 (0%)	964 (<0.1%)	
PhD	0 (0%)	3,801 (<0.1%)	
Unknown	1,775,414	467,515	
income			
≤\$10,000	899,845 (14%)	697,668 (13%)	
\$10,001 - \$30,000	1,086,397 (16%)	1,271,377 (23%)	
\$30,001 - \$50,000	1,124,457 (17%)	1,029,739 (19%)	
\$50,001 - \$80,000	1,410,651 (21%)	986,504 (18%)	
>\$80,000	2,133,521 (32%)	1,443,596 (27%)	
Unknown	2,386,970	882,695	
avg_pmk_cat			
0 TO 4	290,540 (4.2%)	17,642 (17%)	
13 TO 16	516,434 (7.4%)	5,660 (5.5%)	
5 TO 8	2,551,052 (37%)	44,215 (43%)	
9 TO 12	3,575,749 (52%)	36,092 (35%)	